



ATM MANAGEMENT SYSTEM

IN 8086 ASSEMBLY LANGUAGE

Course: Operating System

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Program:
BSCS



Program code:

```
.model small
.stack 100h

.data

pin db '1'
balance db 5
choice db ?
amount db ?

msgTitle db '*** ATM SYSTEM ***$'

msgPin db 0Dh,0Ah,'Enter PIN: $'

msgWrong db 0Dh,0Ah,'Wrong PIN$'

msgMenu db 0Dh,0Ah,'1. Balance'
         db 0Dh,0Ah,'2. Withdraw'
         db 0Dh,0Ah,'3. Deposit'
         db 0Dh,0Ah,'4. Exit'
         db 0Dh,0Ah,'Choose option: $'

msgBal db 0Dh,0Ah,'Current Balance: $'

msgAmt db 0Dh,0Ah,'Enter amount (0-9): $'

msgLow db 0Dh,0Ah,'Insufficient Balance$'

msgBye db 0Dh,0Ah,'Thank You$'

.code

main proc

    ; Initialize data segment
    mov ax, @data
    mov ds, ax

    ; Clear screen
    mov ah, 00h
    mov al, 03h
```

int 10h

; Display title

mov ah, 09h

lea dx, msgTitle

int 21h

; Ask for PIN

mov ah, 09h

lea dx, msgPin

int 21h

; Take PIN input

mov ah, 01h

int 21h

; Compare PIN

cmp al, pin

jne wrong_pin

menu:

; Display menu

mov ah, 09h

lea dx, msgMenu

int 21h

; Take menu choice

mov ah, 01h

int 21h

mov choice, al

; Compare choices

cmp choice, '1'

je show_balance

cmp choice, '2'

je withdraw

cmp choice, '3'

je deposit

```
cmp choice, '4'  
je exit_program
```

```
; Invalid choice  
jmp menu
```

show_balance:

```
; Display balance message  
mov ah, 09h  
lea dx, msgBal  
int 21h
```

```
; Convert number to ASCII  
mov al, balance  
add al, 30h
```

```
; Print balance  
mov dl, al  
mov ah, 02h  
int 21h
```

```
jmp menu
```

withdraw:

```
; Ask amount  
mov ah, 09h  
lea dx, msgAmt  
int 21h
```

```
; Take amount input  
mov ah, 01h  
int 21h
```

```
; Convert ASCII to number  
sub al, 30h
```

```
mov amount, al
```

```
; Check balance  
mov al, balance  
cmp al, amount
```

jb low_balance

; Subtract amount

mov al, balance

sub al, amount

mov balance, al

jmp show_balance

deposit:

; Ask amount

mov ah, 09h

lea dx, msgAmt

int 21h

; Take amount input

mov ah, 01h

int 21h

; Convert ASCII to number

sub al, 30h

; Add into balance

add balance, al

jmp show_balance

low_balance:

; Display insufficient balance

mov ah, 09h

lea dx, msgLow

int 21h

jmp menu

wrong_pin:

; Display wrong PIN message

mov ah, 09h

lea dx, msgWrong

int 21h

exit_program:

; Thank you message

mov ah, 09h

lea dx, msgBye

int 21h

; Exit program

mov ah, 4Ch

int 21h

main endp

end main